

PSP - Global database endpoint for Aurora Postgres

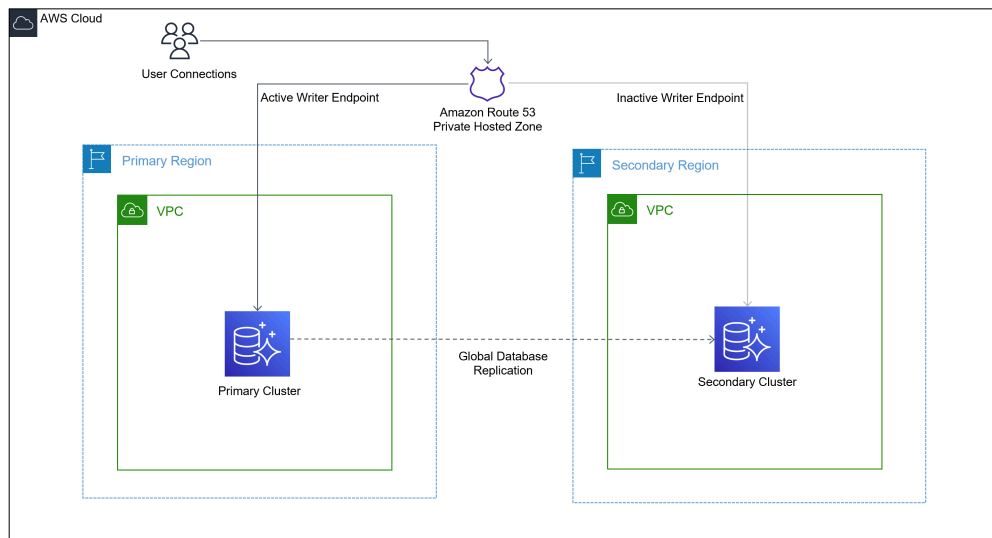
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Overview

In the global database topology, only one Region is primary and all other Regions are secondary. The primary Region contains the only writer instance and also contains the only active writer endpoint. The writer endpoint always points to the active writer node. All secondary Regions also have writer endpoints, but they're inactive.

PSP will be using route 53 CNAME endpoint to connect to the database writer. The automation will take care of updating the CNAME to the writer endpoint of the current primary cluster in the global database in the event of a managed planned failover. There would be change required to current connection configuration and then it would be as below



Prerequisites

- AWS CLI already configured with administrator permission
- The [latest version of Python 3](#)
- The [AWS SDK for Python \(boto3\)](#)
- An AWS account with at least one Aurora global database with at least two Regions
- The [Git command line tools installed](#)

Deployment

- Create cloudformation stack**

```
python3 buildstack.py --template-body 'managed-gdb-cft.yml' --stack-name 'ips-pds-global-endpoint' --consent-anonymous-data-collect 'no' --region-list 'us-east-2,us-west-2'
```

```
Started building stackid 7f667880-4b8c-11ed-99fa-02656e2bc2e4 in Region us-east-2 at: 2022-10-14T06:50:32.276253Z
Started building stackid 80a2c910-4b8c-11ed-8b98-068bb64d3291 in Region us-west-2 at: 2022-10-14T06:50:34.407379Z
Stack creation in progress for region... us-east-2
Stack creation in progress for region... us-west-2
Stack creation in progress for region... us-east-2
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Stack creation in progress for region... us-east-2
Stack creation in progress for region... us-west-2
Stack construction completed for region... us-east-2
Stack creation in progress for region... us-west-2
Stack construction completed for region... us-east-2
```

- Below CF stack will be create in us-west and us-east regions.

	Stack name	Status	Created time	Description
	ips-pds-global-endpoint	 CREATE_COMPLETE	2022-10-14 12:20:32 UTC+0530	Creates supporting resources for automated global database endpoint management solution.

- Create global endpoint for Aurora Postgres Global database

```
python3 create_managed_endpoint.py --cluster-cname-pair '{"psp-test-global":"psp-test-db.psp-ppd.a.intuit.com"}' --hosted-zone-name=psp-ppd.a.intuit.com --region-list 'us-we

Processing region us-west-2 :
Processing regional cluster psp-test :
Hosted Zone psp-ppd.a.intuit.com. already exists. Checking vpcs..
VPC vpc-02eb1ccd0822b6f9e Already exists. checking if CNAME exists..
CNAME psp-test-db.psp-ppd.a.intuit.com doesn't exist. Adding CNAME record..
Cname psp-test-db.psp-ppd.a.intuit.com Successfully created with endpoint psp-test.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com
Added entry to the dynamodb table for cluster psp-test
Processing regional cluster psp-testdr :
No entries made for this cluster. Cluster is not in the current region.

Processing region us-east-2 :
```

- Below will be created and available to use:
 - Global resources created per account:
 - Private hosted zone (Route 53)** – A private hosted zone is created based on the values you passed.
 - CNAME records** – A CNAME record is created inside the hosted zone based on the parameters you passed per cluster.
 - Local resources created per Region:
 - IAM role** – An [AWS Identity and Access Management](#) role is created so the Lambda function can assume this role while running. The role is very granular and minimalistic. It selectively grants only permissions needed to carry out the functions needed to run the Lambda function.
 - Lambda function** – This function is invoked on a global database failover completion event, and updates the CNAME. It contains the core logic of the solution automation.
 - DynamoDB table** – A DynamoDB table named `gdbcnamepair` is created. This table keeps track of the clusters that are managed by this solution.
 - EventBridge rule** – This rule is triggered when a global database completes failover in the Region. This rule has the Lambda function as its target.

The screenshot shows the AWS CloudFormation console for the 'gdb-managed-endpoint' stack. The 'Function overview' tab is active, displaying the configuration for the 'gdb-managed-endpoint' resource. The resource is a Lambda function with the role 'gdb-managed-endpoint-role' and environment variables for 'GDB_ENDPOINT' and 'GDB_REGION'. The function is triggered by an EventBridge rule named 'EventBridge (CloudWatch Events)'. The console also shows a description of the function's purpose and a list of related resources.

- Try to login to your PG Aurora DB using the newly created CNAME

CNAME record before switchover

CNAME record updated after switchover via global endpoint automation

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...	Record ID
☐	psp-ppd.a.intuit.com	NS	Simple	-	No	ns-1536.awsdns-00.co.uk. ns-0.awsdns-00.com. ns-1024.awsdns-00.org. ns-512.awsdns-00.net.	172800	-	-	-
☐	psp-ppd.a.intuit.com	SOA	Simple	-	No	ns-1536.awsdns-00.co.uk. awsdns-hostmaster.amazon.com. 1 7200 900 1209...	900	-	-	-
☐	ppsp-pds-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	ppsp-pds-db.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-	-
☐	psp-test-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	psp-testdr.cluster-cxph5mzesrt.us-east-2.rds.amazonaws.com	1	-	-	-

Reference

- [automate-amazon-aurora-global-database-endpoint-management](#)

[payrolldocs](#)